
DSA0405- FUNDAMENTALS OF DATA SCIENCE

Unit 3 - Solved problems for Class Work

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1. Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.

Answer

Arrange the data:

12, 15, 15, 15, 18, 20, 22, 25

Mean:

$$\begin{aligned} \text{Mean} &= \frac{12 + 15 + 18 + 15 + 20 + 22 + 15 + 25}{8} \\ &= \frac{142}{8} = 17.75 \end{aligned}$$

Median:

$$\text{Median} = \frac{15 + 18}{2} = 16.5$$

Mode:

15 occurs most frequently.

Mode = 15

Range:

$$\text{Range} = 25 - 12 = 13$$

Final Answer

Mean = 17.75, Median = 16.5, Mode = 15, Range = 13

2. Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Answer

Count the occurrences of each value:

Value Frequency

2 3

3 3

4 3

5 2

6 1

Total frequency:

$$3 + 3 + 3 + 2 + 1 = 12$$

Final Answer

The frequency distribution is 2 → 3, 3 → 3, 4 → 3, 5 → 2, 6 → 1.

3. Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

Answer

The data is already arranged in ascending order.

Q2 (Median)

$$Q2 = \frac{15 + 18}{2} = 16.5$$

Q2 = 16.5

Q1

Lower half:

5, 8, 10, 12, 15

Middle value = 10

Q1 = 10

Q3

Upper half:

18, 20, 22, 25, 30

Middle value = 22

Q3 = 22

IQR

$$\begin{aligned} IQR &= Q3 - Q1 \\ &= 22 - 10 = 12 \end{aligned}$$

Final Answer

Q1 = 10, Q2 = 16.5, Q3 = 22, IQR = 12

4. Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.

Answer

Q1

Lower half:

10, 12, 13, 15

$$Q1 = \frac{12 + 13}{2} = 12.5$$

Q3

Upper half:

17, 18, 20, 45

$$Q3 = \frac{18 + 20}{2} = 19$$

IQR

$$IQR = 19 - 12.5 = 6.5$$

Lower Fence

$$\begin{aligned} & Q1 - 1.5(IQR) \\ &= 12.5 - 1.5(6.5) = 2.75 \end{aligned}$$

Upper Fence

$$\begin{aligned} & Q3 + 1.5(IQR) \\ &= 19 + 1.5(6.5) = 28.75 \end{aligned}$$

Any value below 2.75 or above 28.75 is an outlier.

Since:

$$45 > 28.75$$

Final Answer

45 is an outlier.

5. Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.

Answer

Mean

$$\begin{aligned} \text{Mean} &= \frac{5 + 6 + 7 + 7 + 8 + 8 + 9 + 20}{8} \\ &= \frac{70}{8} = 8.75 \end{aligned}$$

Median

$$\text{Median} = \frac{7 + 8}{2} = 7.5$$

Compare:

$$\text{Mean} > \text{Median}$$

$$8.75 > 7.5$$

The extreme value 20 creates a longer tail on the right.

Final Answer

The distribution is positively skewed (right-skewed).

6. A uniformly distributed variable ranges from 10 to 50. Find $P(20 < X < 35)$.

Answer

For a uniform distribution:

$$\begin{aligned} P(a < X < b) &= \frac{b - a}{\text{Maximum} - \text{Minimum}} \\ P(20 < X < 35) &= \frac{35 - 20}{50 - 10} \\ &= \frac{15}{40} = 0.375 \end{aligned}$$

In percentage:

$$0.375 \times 100 = 37.5\%$$

Final Answer

$P(20 < X < 35) = 0.375$ or 37.5%

7. A normal distribution has mean 100 and standard deviation 15. Find the Z-score for $X = 130$.

Answer

Given:

- $X = 130$
- $\mu = 100$
- $\sigma = 15$

Formula:

$$Z = \frac{X - \mu}{\sigma}$$
$$Z = \frac{130 - 100}{15}$$
$$Z = \frac{30}{15} = 2$$

Final Answer

$Z = 2$

Therefore, $X = 130$ is 2 standard deviations above the mean.

8. For the same normal distribution, find $P(X < 130)$.

Answer

From Question 7:

$$Z = 2$$

From the standard normal distribution table:

$$P(Z < 2.00) = 0.9772$$

Therefore:

$$P(X < 130) = 0.9772$$

Percentage:

$$0.9772 \times 100 = 97.72\%$$

Final Answer

$$P(X < 130) = 0.9772 \text{ or } 97.72\%$$

9. Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

Answer

Consider the dataset:

10, 12, 13, 14, 15, 16, 17, 100

Here, 100 is the extreme value.

Before Outlier Treatment

Mean:

$$\text{Mean} = \frac{197}{8} = 24.625$$

Median:

$$\text{Median} = \frac{14 + 15}{2} = 14.5$$

After Removing 100

New dataset:

10, 12, 13, 14, 15, 16, 17

Mean:

$$\text{Mean} = \frac{97}{7} = 13.86$$

Median:

$$\text{Median} = 14$$

Comparison

Measure Before After

Mean 24.625 13.86

Median 14.5 14

Final Answer

The outlier 100 greatly affects the mean, while the median remains almost unchanged. Therefore, the median is more resistant to extreme values.

10. Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

Answer

Consider the student marks:

45, 50, 52, 55, 58, 60, 62, 65, 68, 95

Central Tendency

Mean:

$$\text{Mean} = \frac{610}{10} = 61$$

Median:

$$\text{Median} = \frac{58 + 60}{2} = 59$$

Mode: No mode, since every value occurs only once.

Dispersion

Range:

$$\text{Range} = 95 - 45 = 50$$

Quartiles

Lower half:

45, 50, 52, 55, 58

$$Q1 = 52$$

Upper half:

60, 62, 65, 68, 95

$$Q3 = 65$$

IQR

$$\text{IQR} = 65 - 52 = 13$$

Outlier Detection

Upper fence:

$$\begin{aligned}Q3 + 1.5(IQR) \\= 65 + 1.5(13) = 84.5\end{aligned}$$

Lower fence:

$$\begin{aligned}Q1 - 1.5(IQR) \\= 52 - 1.5(13) = 32.5\end{aligned}$$

Since:

$$95 > 84.5$$

95 is an outlier.

Skewness

Mean = 61

Median = 59

Since:

$$Mean > Median$$

the data is positively/right skewed.

Final Answer

Measure	Value
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Mean	61
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Median	59
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Mode	No Mode
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Range	50
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Q1	52
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Q2	59
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Measure	Value
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Q3	65
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IQR	13
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Outlier	95
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Skewness Positive

Conclusion: The student marks show moderate dispersion, with 95 as an outlier and the distribution being positively skewed.

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